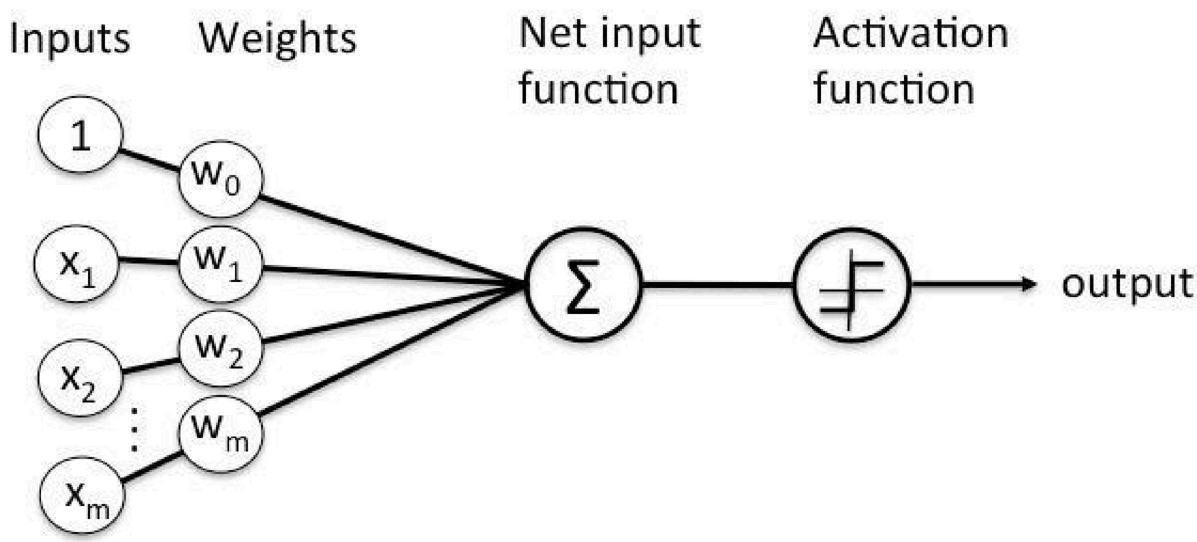
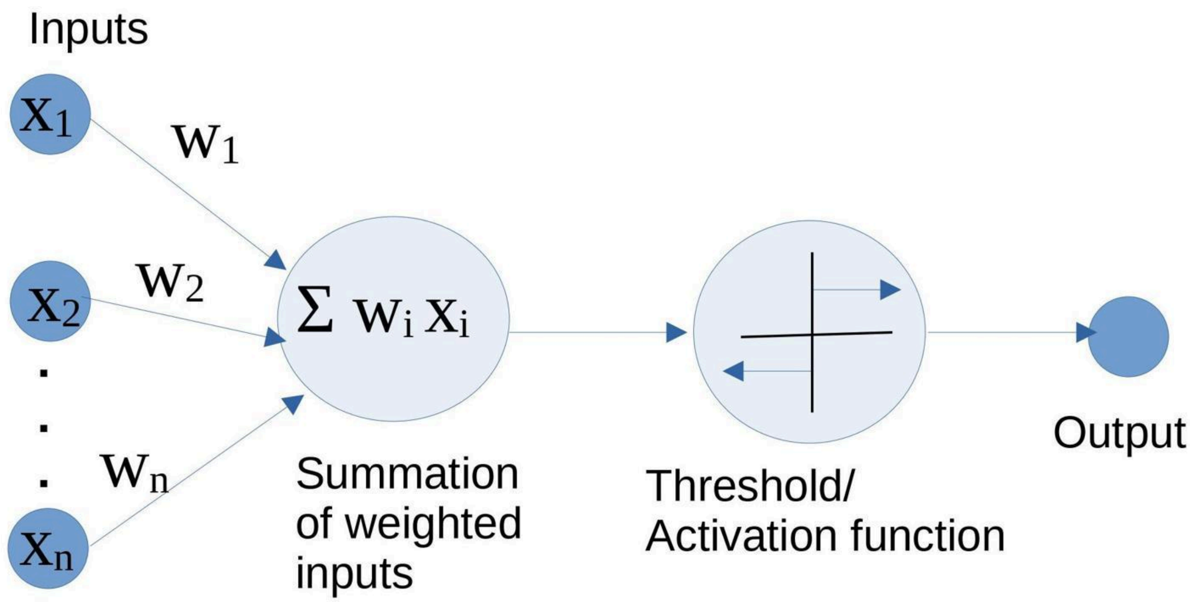
Single Perceptron Diagram Explanation

This diagram shows how a **Single Perceptron** works step by step.



Schematic of Rosenblatt's perceptron.



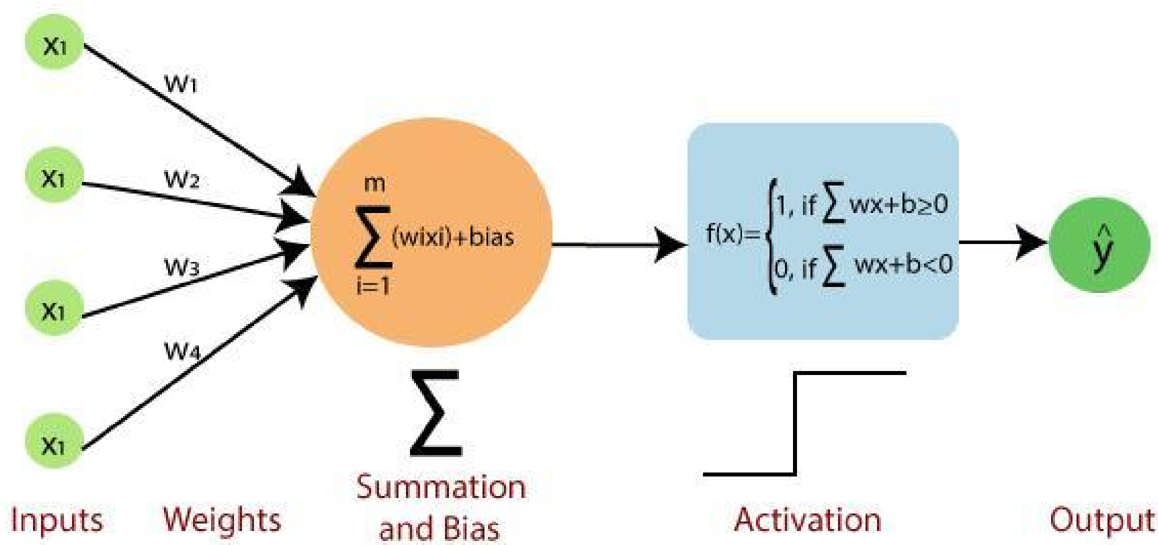


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Diagram Flow

Inputs → Weights → Net Input Function → Activation Function → Output

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1) Inputs (X_1, X_2, X_3 ... X_n)

These are the data values given to the perceptron.

Example:

X1 = Study Hours
X2 = Attendance
X3 = Assignment Score

These inputs enter the network.

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2) Weights (W_0, W_1, W_2 ... W_n)

Weights represent the importance of each input.

- Higher weight = more importance
- Lower weight = less importance

Example:

Study Hours → High importance
Attendance → Medium importance

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3) Bias Input (1 with W_0)

In your diagram, the top input is:

1 → W0

This is called **Bias**.

Bias helps the perceptron make better decisions even when all inputs are zero.

Simple meaning:

Bias = Extra adjustment value

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4) Net Input Function (Σ)

This summation block adds all weighted inputs.

Formula:

Net Input=x1w1+x2w2+x3w3+⋯+xmwm+w0Net\ Input = x_1w_1 + x_2w_2 + x_3w_3 + \cdots + x_mw_m + w_0Net Input=x1w1+x2w2+x3w3+⋯+xmwm+w0

Example:

Study Hours × Weight
+ Attendance × Weight
+ Bias

Everything is added together.

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5) Activation Function

After summation, activation function decides the final answer.

Usually:

If value > threshold → 1
Else → 0

Meaning:

- 1 = YES / PASS
- 0 = NO / FAIL

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6) Output

Final prediction/result comes here.

Example:

Output = PASS

or

Output = FAIL

Full Real-Time Example

Student Pass Prediction

Inputs

Study Hours = 8
Attendance = 90%

Step 1

Inputs go into perceptron.

Step 2

Weights applied.

Step 3

Summation calculates total score.

Step 4

Activation function checks threshold.

Final Output

PASS

Very Simple Analogy

Think of a job interview.

HR checks:

- Communication

- Technical skill
- Experience

Each has different importance (weights).

Finally HR decides:

Selected or Rejected

That decision-making process is similar to a Single Perceptron.

One-Line Summary

Single Perceptron takes inputs, applies weights, calculates total, and gives a simple yes/no output.